AFFORDABLE WI-FI ENABLED FIVE-FINGER GESTURE-CONTROLLED ROBOTIC HAND

Project Category: Product / Technology
Development Stage: Working Prototype
Estimated Project Budget: ₹3,360

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1. Executive Summary

This project presents an affordable five-finger robotic hand designed to reproduce human hand movements using gesture tracking and wireless communication. It combines computer vision, microcontrollers, servo motors and a homemade mechanical structure to demonstrate natural human-robot interaction.

The system captures human hand gestures through a camera and translates the detected finger movements into wireless control commands. An ESP8266 NodeMCU and servo motors control the robotic fingers, allowing them to reproduce corresponding movements.

The project explores the potential of affordable robotic technology in assistive applications, remote manipulation, education and research.

2. Problem Statement

People with limited hand mobility due to physical disabilities, injuries or other conditions may experience difficulties performing everyday activities involving gripping, holding and manipulating objects.

Advanced robotic and prosthetic hand technologies can be expensive and may involve complicated control systems. This creates accessibility challenges, particularly for people and communities with limited financial resources.

There is a need to explore affordable and intuitive robotic systems that can translate natural human hand movements into mechanical movements.

3. Proposed Solution

Our proposed solution is a low-cost, Wi-Fi-enabled five-finger robotic hand that mirrors human finger movements through gesture tracking.

The system combines camera-based hand tracking, wireless communication and servo motor control. Its mechanical structure is constructed using affordable materials such as cardboard, threads and rubber bands.

The project demonstrates how commonly available electronic components can be integrated to create an affordable robotic manipulation system.

4. How the Solution Works

A camera captures the user's hand movements, while computer vision software detects individual finger positions and movements.

The detected movements are converted into control commands and transmitted wirelessly over Wi-Fi. An ESP8266 NodeMCU receives these commands and controls five servo motors, each responsible for one robotic finger.

The servo motors pull threads to bend the fingers, while rubber bands help return them to their original positions.

This enables the robotic hand to reproduce corresponding human finger movements without requiring a direct wired connection between the user and the robotic hand.

5. Hardware and Components

Component| Quantity
ESP8266 NodeMCU| 2
Arduino Uno| 2
Servo motors| 5
Phone chargers| 3
USB cables| 3

Threads and rubber bands| As required
Cardboard and supporting materials| As required
Connecting wires| As required

6. Software and Technologies

- Computer vision-based hand gesture tracking.
- ESP8266 wireless communication.
- Arduino-based microcontroller programming.
- Servo motor control.
- HTML-based browser control interface.
- Wi-Fi communication between the control system and robotic hand.

7. Prototype Development

The project has reached the working prototype stage.

The robotic hand has been mechanically assembled using affordable materials. Five servo motors are used to control individual fingers, and the ESP8266 NodeMCU provides wireless control.

The browser-based control interface has been tested, and the robotic fingers respond to wireless commands.

Further refinement is planned to improve movement accuracy, synchronization and gesture-tracking performance.

8. Innovation and Uniqueness

The main innovation is combining five-finger robotic movement with natural human gesture tracking and wireless communication.

Key features include:

- Independent control of five robotic fingers.
- Wireless operation through Wi-Fi.

- Gesture-based human-robot interaction.
- Affordable and locally available construction materials.
- A homemade mechanical finger mechanism.
- Expandable design for future assistive and remote-operation applications.

9. Target Beneficiaries

The primary potential beneficiaries are people with limited hand mobility due to disabilities, injuries or physical conditions.

Other potential beneficiaries include rehabilitation centres, healthcare institutions, assistive technology developers, engineering students, research laboratories and industries interested in remote manipulation.

10. Social Impact

The project explores the possibility of making basic robotic manipulation technology more affordable and accessible.

It could support future developments in assistive technology, rehabilitation research and remote handling systems. It also encourages students and local innovators to explore practical robotics using affordable components.

The current prototype demonstrates the underlying technology but is not yet a medically validated prosthetic device.

11. Applications

Assistive Technology: Future development could help people with limited hand mobility perform certain gripping activities.

Remote Manipulation: The system could be adapted to control robotic fingers remotely.

Education and Research: Schools and engineering institutions can use the project to explore robotics, microcontrollers and human-computer interaction.

Industrial Applications: Future adaptations could support remote object handling in selected environments.

12. Estimated Project Budget

Component| Quantity| Estimated Cost
ESP8266 NodeMCU| 2| ₹800
Arduino Uno| 2| ₹960
Servo motors| 5| ₹900
Phone chargers| 3| ₹300
USB cables| 3| ₹150
Wires and construction materials| -| ₹250
Total| | ₹3,360

Note: Some accessory prices are estimates and should be replaced with actual purchase prices where available.

13. Beneficiary Adoption and Localization

People with limited hand mobility, rehabilitation centres, hospitals, educational institutions and industries could adopt or further develop this solution.

It can be localized using affordable components available in local markets, collaboration with nearby engineering colleges and healthcare centres, regional-language instructions such as Tamil, and customized designs suited to local needs and budgets.

14. Scalability and Global Expansion

The solution could be expanded by improving its mechanical structure, movement accuracy, durability and control systems.

Future versions could use lightweight and stronger materials, improved gesture tracking and personalized controls.

Financial support, technical mentorship, research partnerships, product testing, manufacturing assistance and collaboration with assistive technology organizations

would help develop the project into a more reliable and accessible system for wider use.

15. Future Development

- Improve finger movement precision and synchronization.
- Develop a stronger and lighter mechanical structure.
- Improve real-time gesture tracking.
- Explore customizable control methods.
- Conduct usability testing with appropriate supervision.
- Investigate safe assistive applications.
- Explore affordable manufacturing and wider distribution.

16. Current Limitations

The current prototype is an experimental demonstration of gesture-controlled robotic movement.

It has not yet been clinically tested or certified for use as a prosthetic device. Further work is required to improve mechanical durability, user safety, grip performance and practical usability.

17. Conclusion

This project demonstrates how affordable electronics, computer vision, wireless communication and mechanical engineering can be combined to develop a five-finger robotic hand.

By focusing on affordability, natural gesture-based control and future assistive applications, the project provides a foundation for further development in accessible robotics and human-robot interaction.

The long-term vision is to contribute to affordable robotic technologies that can improve accessibility and expand opportunities for people with limited hand mobility.